

NETFLIX *Business Case Study*

~ by: Shivaling

Email: shivalingb09@gmail.com

- **About NETFLIX**

Netflix is one of the most popular media and video streaming platforms. They have over 10000 movies or tv shows available on their platform, as of mid-2021, they have over 222M Subscribers globally. This tabular dataset consists of listings of all the movies and tv shows available on Netflix, along with details such as - cast, directors, ratings, release year, duration, etc.

- **Business Problem**

Analyze the data and generate insights that could help Netflix in deciding which type of shows/movies to produce and how they can grow the business in different countries

- **Dataset**

Link: [Dataset link](#)

(After clicking on the above link, you can download the files by right-clicking on the page and clicking on "Save As", then naming the file as per your wish, with .csv as the extension.)

The dataset provided to you consists of a list of all the TV shows/movies available on Netflix:

Show_id: Unique ID for every Movie / Tv Show

Type: Identifier - A Movie or TV Show

Title: Title of the Movie / Tv Show

Director: Director of the Movie

Cast: Actors involved in the movie/show

Country: Country where the movie/show was produced

Date_added: Date it was added on Netflix

Release_year: Actual Release year of the movie/show

Rating: TV Rating of the movie/show

Duration: Total Duration - in minutes or number of seasons

Listed_in: Genre

Description: The summary description

- **Google Colab Notebook**

This report further contains the Google Colab Notebook converted used for Data Analysis.

The Link for Google Colab Notebook:

<https://colab.research.google.com/drive/11hT8Miipg3ARq15KD2Xdq72laj0PYUqU?usp=sharing>

- The following pages are the Google Colab Notebook.

NETFLIX Business Case Study

- Netflix is one of the most popular media and video streaming platforms. They have over 10000 movies or tv shows available on their platform, as of mid-2021, they have over 222M Subscribers globally. This tabular dataset consists of listings of all the movies and tv shows available on Netflix, along with details such as - cast, directors, ratings, release year, duration, etc.

1 # Importing the required Python Libraries

1 import numpy as np
2 import pandas as pd
3 import matplotlib.pyplot as plt
4 import seaborn as sns

1 # Load the Netflix Dataset to be analysed

1 !gdown 1XWeptwqELLr0_7eZt4oLqhzHmb7TfmBp -O netflix.csv

Downloading...
From: https://drive.google.com/uc?id=1XWeptwqELLr0_7eZt4oLqhzHmb7TfmBp
To: /content/netflix.csv
100% 3.40M/3.40M [00:00<00:00, 148MB/s]

1 # Display first few rows of the dataset

1 netflix = pd.read_csv('netflix.csv')
2 netflix.head(2)

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
					Ama Qamata, ...						International TV ...	After crossing ...

Next steps: [Generate code with netflix](#) [New interactive sheet](#)

1 Start coding or generate with AI.

1 Start coding or generate with AI.

Understanding the Dataset: structure, shape, data types, information, columns

1 netflix.dtypes

show_id	object
type	object
title	object
director	object
cast	object
country	object
date_added	object
release_year	int64
rating	object
duration	object
listed_in	object
description	object

dtype: object

1 netflix.shape

(8807, 12)

1 netflix.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   show_id         8807 non-null   object
1   type            8807 non-null   object
2   title           8807 non-null   object
3   director        6173 non-null   object
4   cast            7982 non-null   object
5   country         7976 non-null   object
6   date_added      8797 non-null   object
7   release_year    8807 non-null   int64
8   rating          8803 non-null   object
9   duration        8804 non-null   object
10  listed_in       8807 non-null   object
11  description      8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

1 # There are 6 columns, which are completely Non Null and other 6 have some cells which are Null/Na.

1 netflix.describe(include='all')

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description	
count	8807	8807	8807	6173	7982	7976	8797	8807.000000	8803	8804	8807	8807	
unique	8807	2	8807	4528	7692	748	1767	NaN	17	220	514	8775	
top	s8807	Movie	Zubaan	Rajiv Chilaka	David Attenborough	United States	January 1, 2020	NaN	TV-MA	1 Season	Dramas, International Movies	Paranormal activity at a lush, abandoned prope...	
freq	1	6131	1	19	19	2818	109	NaN	3207	1793	362	4	
mean	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2014.180198	NaN	NaN	NaN	NaN	
std	NaN	NaN	NaN	NaN	NaN	NaN	NaN	8.819312	NaN	NaN	NaN	NaN	
min	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1925.000000	NaN	NaN	NaN	NaN	
25%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2013.000000	NaN	NaN	NaN	NaN	
50%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2017.000000	NaN	NaN	NaN	NaN	

1 # From the above analysis it is evident the Netflix Platform currently has movies and tv shows which released in years ranging from 1

1 netflix.nunique()

```
0
show_id    8807
type        2
title      8807
director   4528
cast       7692
country    748
date_added 1767
release_year 74
rating      17
duration    220
listed_in   514
description 8775
dtype: int64
```

1 # from the above information, we can deduce Netflix platform currently provides content which were released in 74 different years
2 # There are total 8807 no. of movies/tv shows available to watch on Netflix as of now

1 netflix.duplicated().sum()

```
np.int64(0)
```

1 # It is confirmed that in dataset, there are no duplicate entries or rows.

Problem Statement

- Analyze the data and generate insights that could help Netflix in deciding which type of shows/movies to produce and how they can grow the business in different countries

Some Exploratory Problems:

- What type of content is available in different countries?
- How has the number of movies released per year changed over the last 20-30 years?
- Comparison of tv shows vs. movies.
- What is the best time to launch a TV show?
- Analysis of actors/directors of different types of shows/movies.
- Does Netflix has more focus on TV Shows than movies in recent years
- Understanding what content is available in different countries

Handling the missing Values

```
1 netflix.isna().sum()

      0
show_id  0
type     0
title    0
director 2634
cast     825
country  831
date_added 10
release_year  0
rating     4
duration   3
listed_in  0
description 0

dtype: int64
```

```
1 # Handle the 'date_added' column for it null values, and convert the potential date columns into date_time
```

```
1 netflix['date_added'] = pd.to_datetime(netflix['date_added'], errors='coerce')
2
```

```
1 netflix['date_added'].isnull().sum()

np.int64(98)
```

```
1 # All the null values in 'date_added' column have been dealt with and has been replaced with Nat, i.e Not a Time.
```

```
1 # Now, for all other columns which have null entries and string types, replacing them with 'Unknown'
```

```
1 null_col = ['director', 'cast', 'country', 'rating', 'duration']
2 for col in null_col:
3     netflix[col].fillna('Unknown', inplace=True)

/tmp/ipykernel_1554/2874067278.py:3: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignm
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values
For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].metho

netflix[col].fillna('Unknown', inplace=True)
```

```
1 netflix.isna().sum()
```

```

      0
show_id    0
type       0
title      0
director   0
cast       0
country    0
date_added 98
release_year 0
rating     0
duration   0
listed_in  0
description 0
dtype: int64

```

```
1 # All null values have been dealt with, accordingly.
```

Conversion of Categorical Columns

```

1 cat_cols = ['type', 'director', 'cast', 'country', 'rating', 'listed_in' ]
2 for col in cat_cols:
3     netflix[col] = netflix[col].astype('category')

```

Extraction of month and year of date_added

```

1 netflix['month_added'] = netflix['date_added'].dt.month_name()
2 netflix['year_added'] = netflix['date_added'].dt.year

```

▼ Data Analysis

```
1 # Movies v/s TV Shows
```

```
1 netflix['type'].value_counts()
```

```

      count
type
Movie    6131
TV Show   2676

```

```
dtype: int64
```

```
1 # Netflix is dominated by Movie Contents, alsmot 2.5 times more no. of movies available than that of TV Shows.
```

```

1 # Dfiffrent Ratings and their counts
2
3 netflix['rating'].value_counts()

```

	count
rating	
TV-MA	3207
TV-14	2160
TV-PG	863
R	799
PG-13	490
TV-Y7	334
TV-Y	307
PG	287
TV-G	220
NR	80
G	41
TV-Y7-FV	6
Unknown	4
NC-17	3
UR	3
66 min	1
74 min	1
84 min	1

dtype: int64

```
1 # Different Years of content available on Netflix based on their release Year, sorted by Year latest first
2
3 netflix['release_year'].value_counts().sort_index(ascending = False)
```

	count
release_year	
2021	592
2020	953
2019	1030
2018	1147
2017	1032
...	...
1945	4
1944	3
1943	3
1942	2
1925	1

74 rows × 1 columns

dtype: int64

```
1 # Analysis of Countries in which the content was produced, Unique Countries
2
3 netflix['country'].nunique()
```

749

```
1 # Top Countries Producing contents
2
3 netflix['country'].value_counts().head(10)
```

	count
country	
United States	2818
India	972
Unknown	831
United Kingdom	419
Japan	245
South Korea	199
Canada	181
Spain	145
France	124
Mexico	110

dtype: int64

1 # US, India. Japan and UK dominate in producing the content. There are multiple contents which are produced in more than 1 country.

```
1 # Top Genres of content available
2
3 netflix['listed_in'].value_counts().head(10)
```

	count
listed_in	
Dramas, International Movies	362
Documentaries	359
Stand-Up Comedy	334
Comedies, Dramas, International Movies	274
Dramas, Independent Movies, International Movies	252
Kids' TV	220
Children & Family Movies	215
Children & Family Movies, Comedies	201
Documentaries, International Movies	186
Dramas, International Movies, Romantic Movies	180

dtype: int64

1 # Dramas, International Movies, Documentaries, Stand-Up Comedy, dominate the collections.

Categorical Non - Graphical analysis

Data Processing is required for following columns:

- cast
- director
- country
- listed_in

They contain multiple categories within themselves in each entry.

Data Pre Processing

```
1 # Individual Countrys for a movie/tv show
2 country_n = netflix.assign(country = netflix['country'].str.split(',')).explode('country')
3 country_n['country'] = country_n['country'].str.strip()
4
5 # Individual Cast for a movie/tv show
6 cast_n = netflix.assign(cast = netflix['cast'].str.split(',')).explode('cast')
7 cast_n['cast'] = cast_n['cast'].str.strip()
8
9 # Individual Director for a movie/tv show
10 director_n = netflix.assign(director = netflix['director'].str.split(',')).explode('director')
11 director_n['director'] = director_n['director'].str.strip()
12
13 # Individual listed_in genres for a movie/tv show
```

```
1 # No. of different Countries content on Netflix
2
3 country_n['country'].nunique()
```

124

```
1 # Top Countries prducing content, available on Netflix
2
3 country_n['country'].value_counts().head(10)
```

	count
country	
United States	3690
India	1046
Unknown	831
United Kingdom	806
Canada	445
France	393
Japan	318
Spain	232
South Korea	231
Germany	226

dtype: int64

```
1 # Netflix hosts content originating from 124 unique countries, indicating a highly diverse global content library.
2 # This helps undesrtand the Global wide reach of Netflix to its consumers.
3 # The above analysis show the no. of movies/shows produced in respective countries excatly throughout the available content on Netfl
4 # Clearly US, India, UK, Canada, Framce, dominates the charts, and hence further investments and promotions can be done in these co
5 # It is in contradiction to analysis of no. of unique country before doing the separation.
```

```
1 cast_n['cast'].nunique()
```

36440

```
1 cast_n['cast'].value_counts().head(10)
```

	count
cast	
Unknown	825
Anupam Kher	43
Shah Rukh Khan	35
Julie Tejjwani	33
Naseeruddin Shah	32
Takahiro Sakurai	32
Rupa Bhimani	31
Om Puri	30
Akshay Kumar	30
Yuki Kaji	29

dtype: int64

```
1 Netflix has content for 36440 different Cast, i.e. Actors and actress in their available movies and tv shows (including the Unknown
2 This analysis helps to understand consumers interest regarding actors based content cosumption.
3 Excluding the Unknown Cast, Anupam Kher is most featured Actor on Netflix followed by Shah Rukh Khan and Julie Tejjwani
```

```
1 director_n['director'].nunique()
```

4994

```
1 director_n['director'].value_counts().head(10)
```


	count
director	
Unknown	2634
Rajiv Chilaka	22
Jan Suter	21
Raúl Campos	19
Suhas Kadav	16
Marcus Raboy	16
Jay Karas	15
Cathy Garcia-Molina	13
Martin Scorsese	12
Youssef Chahine	12

dtype: int64

```
1 # Netflix has provided platform for contents directed by 4994 different directors from all over the world (including the Unknowns)
2 # This attracts the cinephile consumers to get their favourite directors work to watch
3 # Excluding Unknowns, Rajiv Chilaka is most featured director on Netflix followed by Jan Sutar, Raul Campos
```

```
1 listed_in_n['listed_in'].nunique()
```

42

```
1 listed_in_n['listed_in'].value_counts().head(10)
```

	count
listed_in	
International Movies	2752
Dramas	2427
Comedies	1674
International TV Shows	1351
Documentaries	869
Action & Adventure	859
TV Dramas	763
Independent Movies	756
Children & Family Movies	641
Romantic Movies	616

dtype: int64

```
1 listed_in_n[listed_in_n['type'] == 'Movie']['listed_in'].value_counts().head(10)
```

	count
listed_in	
International Movies	2752
Dramas	2427
Comedies	1674
Documentaries	869
Action & Adventure	859
Independent Movies	756
Children & Family Movies	641
Romantic Movies	616
Thrillers	577
Music & Musicals	375

dtype: int64

```
1 listed_in_n[listed_in_n['type'] == 'TV Show']['listed_in'].value_counts().head(10)
```

	count
listed_in	
International TV Shows	1351
TV Dramas	763
TV Comedies	581
Crime TV Shows	470
Kids' TV	451
Docuseries	395
Romantic TV Shows	370
Reality TV	255
British TV Shows	253
Anime Series	176

dtype: int64

```
1 # Netflix has International Movies most with Dramas followed by Comedies and Documentaries listed in Genres.
2 # Among the TV Shows, International TV Shows, Dramas, Comedies, and Crimes genres dominate the Netflix.
3 # Total of 42 genres of content is available for consumers on Netflix.
4 # This shows the wide variety coverage by Netflix for their consumers
```

```
1 # Splitting these columns will help to analyse Country wise Content availability based on cast, director, listed_in genres, individu
```

```
1 # Understanding what content is available in different countries
2 country_n[country_n['type'] == 'Movie']['country'].value_counts().head(10)
```

	count
country	
United States	2752
India	962
United Kingdom	534
Unknown	440
Canada	319
France	303
Germany	182
Spain	171
Japan	119
China	114

dtype: int64

```
1 # US leads with most no. of Movies produced there among the available Movies on Netflix, followed by India, UK, Canada, France, Germa
2 # Japan, China being in Top 10 Countries producing content on Netflix.
```

```
1 country_n[country_n['type'] == 'TV Show']['country'].value_counts().head(10)
```

	count
country	
United States	938
Unknown	391
United Kingdom	272
Japan	199
South Korea	170
Canada	126
France	90
India	84
Taiwan	70
Australia	66

dtype: int64

```
1 # US leads even in the TV Show type of Content, followed by UK, Japan, South Korea, Canada, France, India,
2 # Taiwan, Australia among the Top 10 TV Show producers on Netflix
```

```
1 cast_n[cast_n['type'] == 'Movie']['cast'].value_counts().head(10)
```

count	
cast	
Unknown	475
Anupam Kher	42
Shah Rukh Khan	35
Naseeruddin Shah	32
Om Puri	30
Akshay Kumar	30
Amitabh Bachchan	28
Paresh Rawal	28
Julie Tejwani	28
Rupa Bhimani	27

dtype: int64

```
1 cast_n[cast_n['type'] == 'TV Show']['cast'].value_counts().head(10)
```

count	
cast	
Unknown	350
Takahiro Sakurai	25
Yuki Kaji	19
Daisuke Ono	17
Junichi Suwabe	17
Ai Kayano	17
Yuichi Nakamura	16
Jun Fukuyama	15
Yoshimasa Hosoya	15
David Attenborough	14

dtype: int64

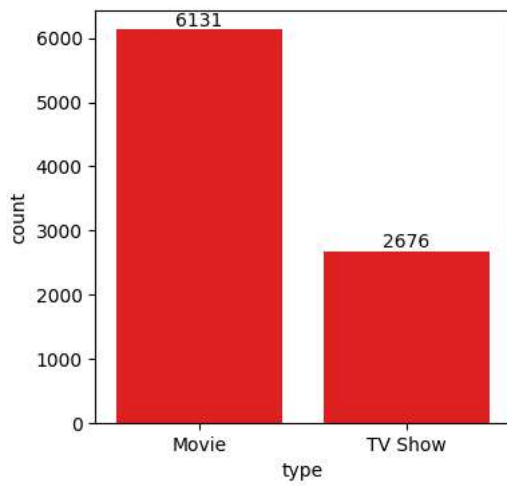
```
1 # Excluding the Unknowns, In Movies, Anupam Kher and In TV Shows, Takahiro Sakurai are most featured on Netflix.
```

```
1 Start coding or generate with AI.
```

Data Visualization

Univariate Analysis

```
1 # Movies v/s TV Show
2
3 plt.figure(figsize = (4,4))
4 mov_tv = sns.countplot(data = netflix, x = 'type', color = 'r')
5
6 for container in mov_tv.containers:
7     mov_tv.bar_label(container)
8
9 plt.show()
```



1 # Movies dominate the Netflix Collection with 6131 no. of movies and 2676 no. of TV Shows.

1 netflix.head(1)

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description	month_added
0	s1	Movie	Dick Johnson	Kirsten	Unknown	United States	2021-09-25	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, she reflects on his life and the impact he has had on her.	September

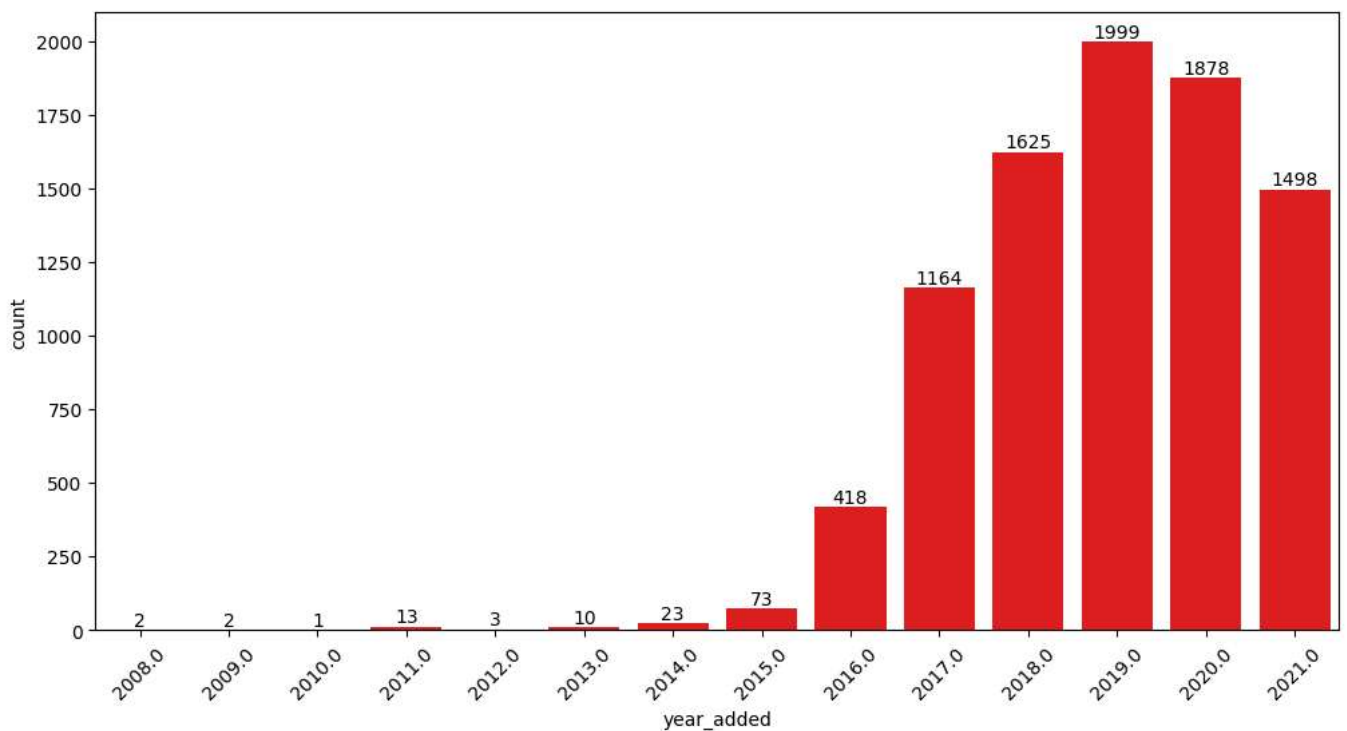
Next steps: [Generate code with netflix](#) [New interactive sheet](#)

1 netflix['year_added'].max(), netflix['year_added'].min()

(2021.0, 2008.0)

```

1 # Content made available every year
2
3 plt.figure(figsize = (12, 6))
4
5 fig = sns.countplot(data = netflix, x = 'year_added', color = 'r')
6 for container in fig.containers:
7     fig.bar_label(container)
8
9 plt.xticks(rotation = 45)
10 plt.show()
```



1 # In Year 2019, Netflix added most no. of contents, 2016 no. of titles, and least was in Year 2010.

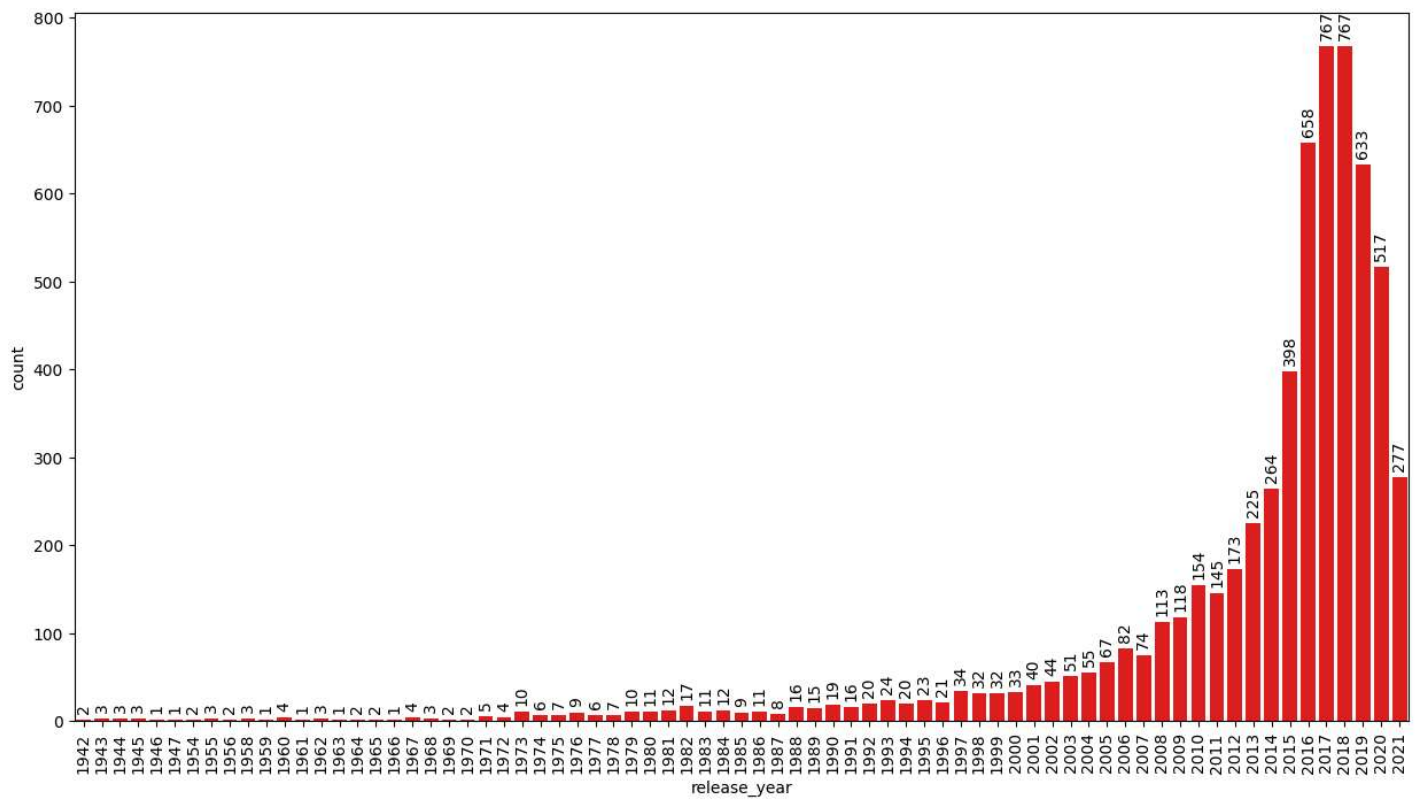
```

1 # Movies released every year
2
3 plt.figure(figsize = (15, 8))
4
5 ry = sns.countplot(data = netflix[netflix['type'] == 'Movie'], x = 'release_year', color = 'red')
```

```

6 for container in ry.containers:
7     ry.bar_label(container, rotation = 90, padding = 4)
8
9 plt.xticks(rotation = 90)
10 plt.show()

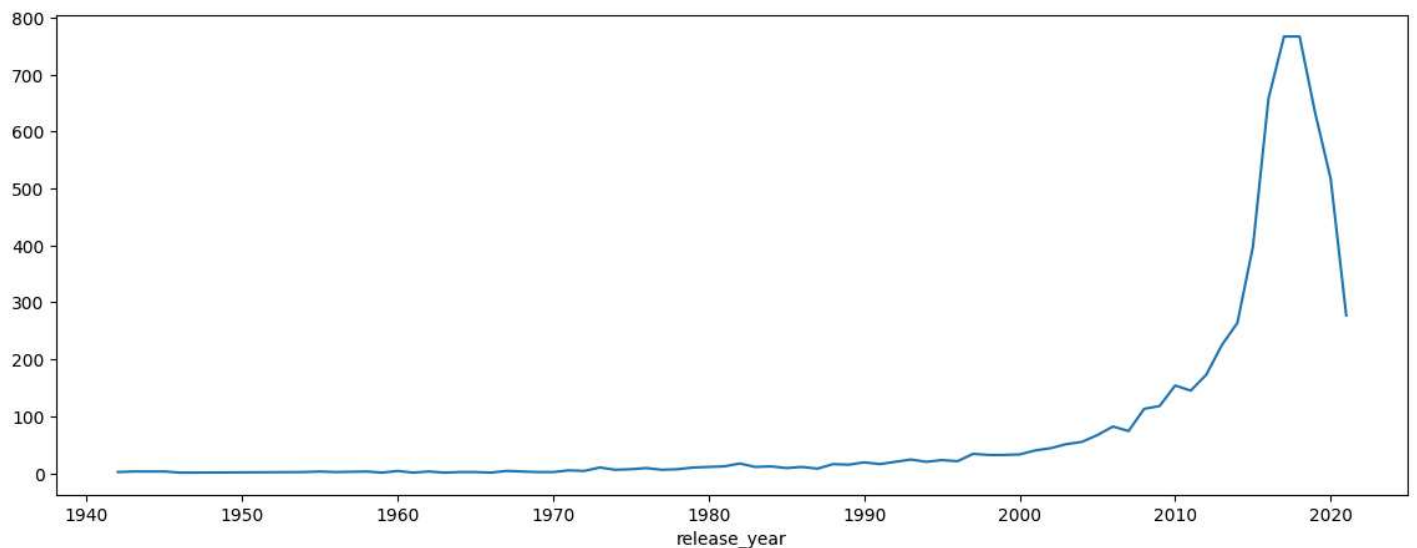
```



```

1 movie_df = netflix[netflix['type']=="Movie"]
2
3 movie_df['release_year'].value_counts().sort_index().plot(figsize=(14,5))
4 plt.show()

```



```

1 # Rating Distribution
2
3 netflix['rating'].nunique()

```

18

```

1 netflix['rating'].value_counts()

```

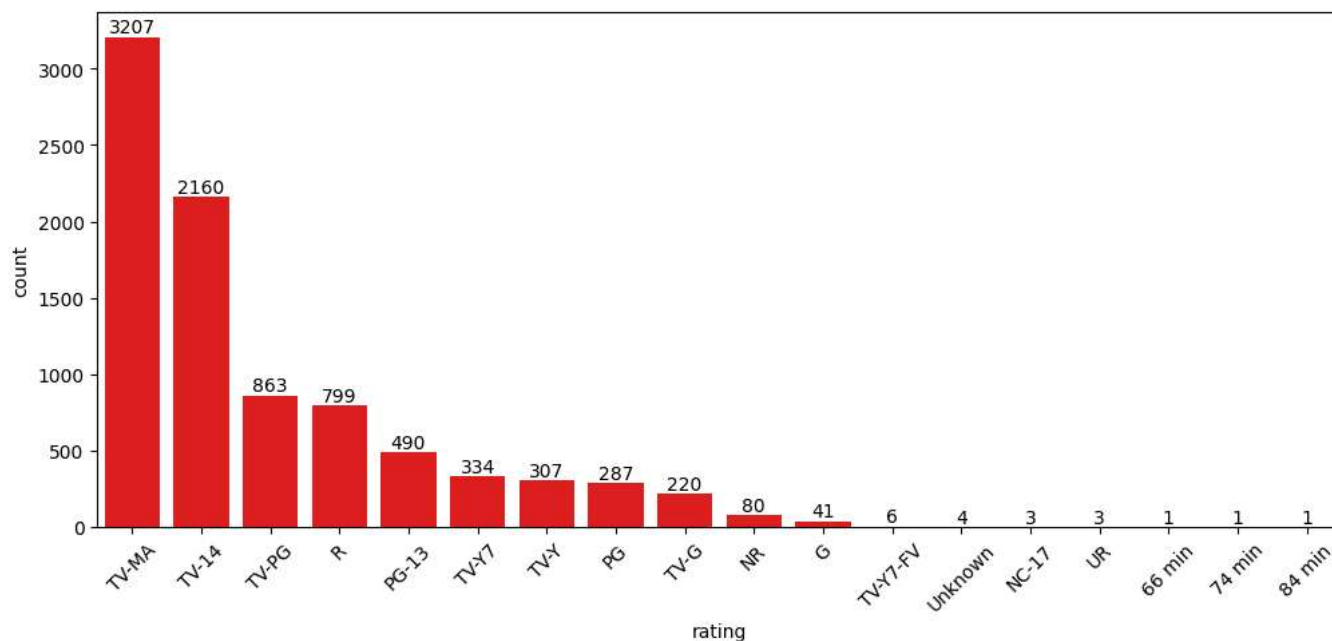
	count
rating	
TV-MA	3207
TV-14	2160
TV-PG	863
R	799
PG-13	490
TV-Y7	334
TV-Y	307
PG	287
TV-G	220
NR	80
G	41
TV-Y7-FV	6
Unknown	4
NC-17	3
UR	3
66 min	1
74 min	1
84 min	1

dtype: int64

```

1 plt.figure(figsize = (12, 5))
2 rd = sns.countplot(data = netflix, x = 'rating', order = netflix['rating'].value_counts().index, color = 'r')
3 for container in rd.containers:
4     rd.bar_label(container)
5
6 plt.xticks(rotation = 45)
7 plt.show()

```



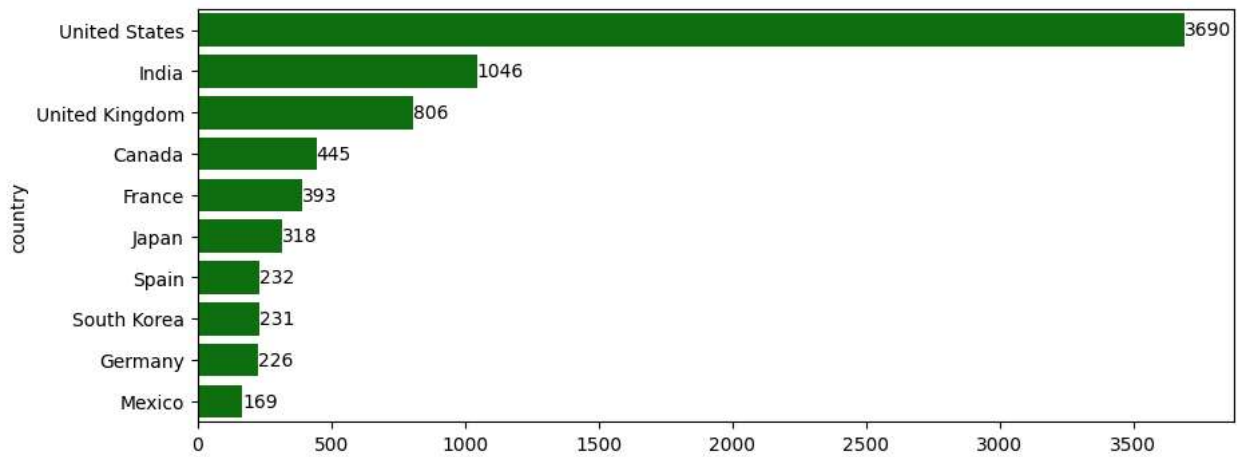
1 # Content Rated as TV-MA tops the charts for Ratings, means Netflix content attracts mostly adults above age 17,
 2 # Ratings are then followed by TV-14 and TV-PG, i.e. contents attract teenagers and families where parental guidance is needed

Data Visualisation using the preprocessed columns

```

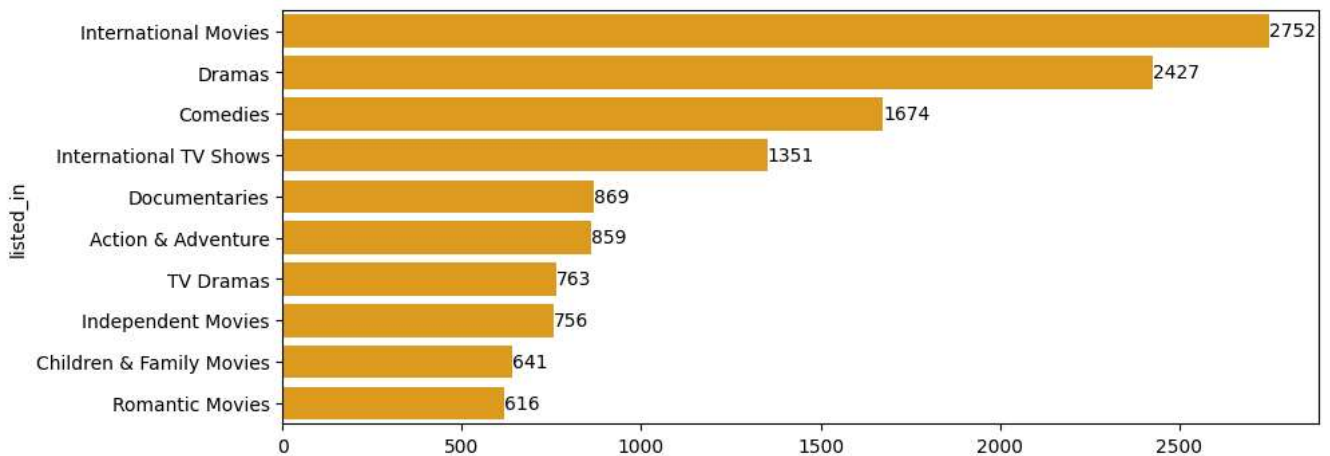
1 # Top Countries
2
3 plt.figure(figsize = (10, 4))
4
5 top_c = country_n[country_n['country'] != 'Unknown']['country'].value_counts().head(10)
6
7 tc = sns.barplot( x = top_c.values, y = top_c.index, color = 'green')
8 for container in tc.containers:
9     tc.bar_label(container)

```



1 # The United States dominates Netflix's catalogue, indicating Netflix's strong dependence on Hollywood productions,
 2 # while simultaneously expanding international content in countries such as India, Japan and South Korea,
 3 # followed by India, United Kingdom, Canada, France
 4 # Investments should be made in these Top countries to maintain and grow consumer attraction by providing more variety of content,
 5 # For under performing Countries, promotions, auditions, advertisements campaigns can be run to attract directors, cast, consumer.

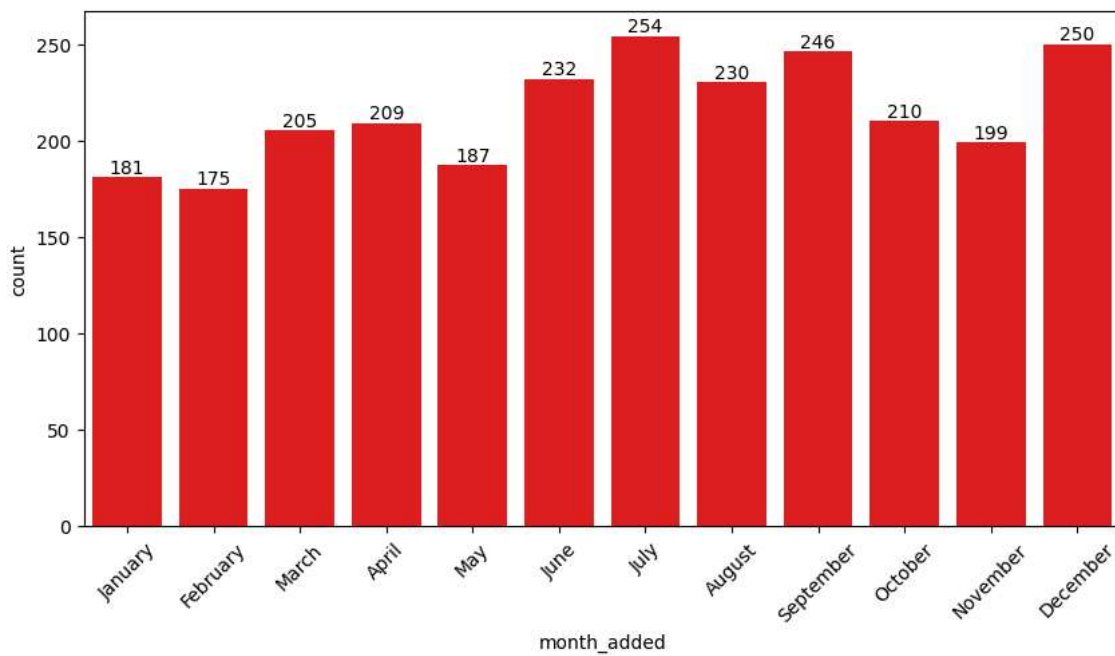
```
1 # Top Genres
2
3 plt.figure(figsize = (10, 4))
4
5 genres = listed_in_n['listed_in'].value_counts().head(10)
6
7 tg = sns.barplot(x = genres.values, y = genres.index, color = 'orange')
8 for container in tg.containers:
9     tg.bar_label(container)
```



1 # Currently Netflix provides most no. of International Movies Genre content for its consumers, followed by Drama, Comedies, TV Shows,
 2 # Documentaries, etc. TV Shows and TV Dramas are in Top 10 which reflects, they have popular among viewers in recent times.

1 # What is the best time to launch a TV show?
 2 # Visualisation by Month added

```
1 order = ["January", "February", "March", "April", "May", "June", "July", "August", "September", "October", "November", "December"]
2
3 plt.figure(figsize = (10, 5))
4
5 ma = sns.countplot(data = netflix[netflix['type'] == 'TV Show'], x = 'month_added', order = order, color = 'r')
6 for container in ma.containers:
7     ma.bar_label(container)
8
9 plt.xticks(rotation = 45)
10 plt.show()
```

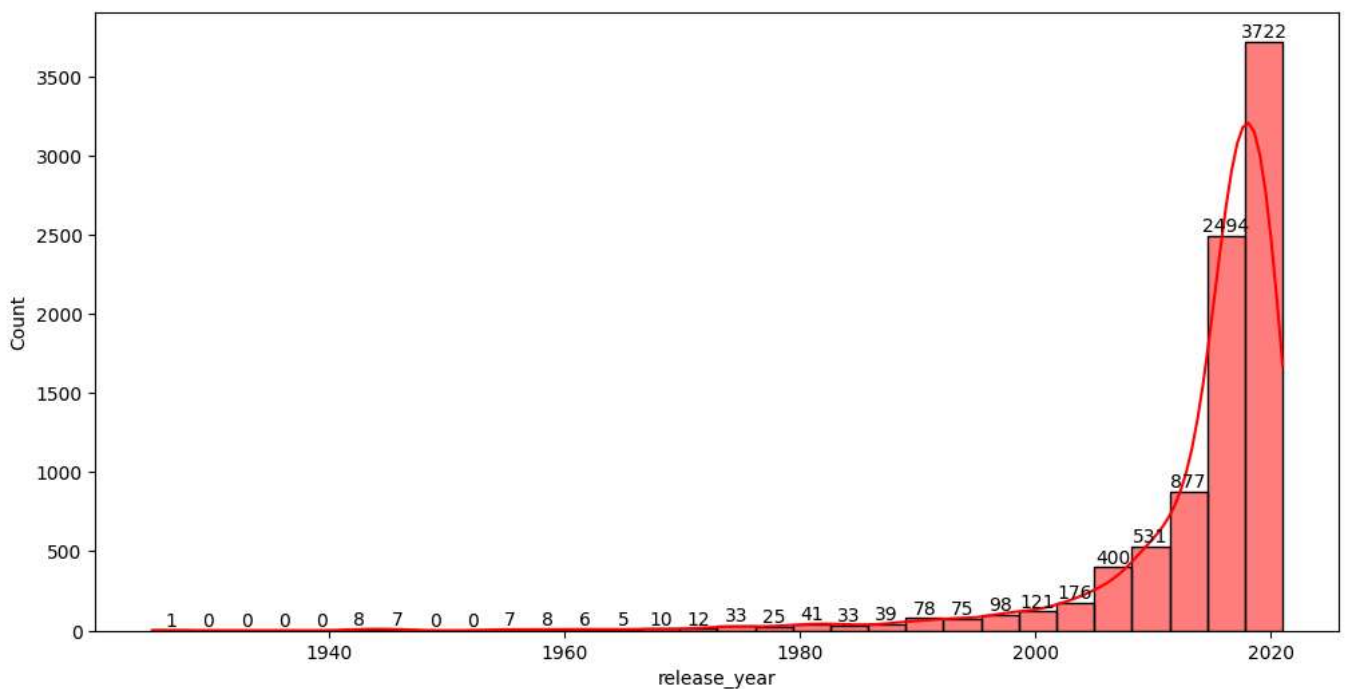


- 1 # Netflix adds the TV Shows on to the platform when they are released, hence the best time to launch a TV Show is in month of July
- 2 # According to analysis, TV shows have been added the least in the month of February.
- 3 # The above charts shows TV Shows are being released most during the Year End winter Season, and least during start of New Year

```

1 # Yearwise Release Distribution
2
3 plt.figure(figsize = (12, 6))
4
5 yr = sns.histplot(netflix['release_year'], bins = 30, kde = True, color = 'r')
6 for container in yr.containers:
7     yr.bar_label(container)
8
9 plt.show()

```



- 1 # Netflix has mostly content available, which are released after 2000,
- 2 # and most no. of content being released during 2020-2021 with 3722 no. of titles being released.
- 3 # Netflix's library is heavily concentrated on titles released after 2000, indicating a strategic focus on contemporary content rather than older titles.

```

1 # Duration of Movies Analysis
2
3 movies = netflix[(netflix['type'] == 'Movie') & (netflix['duration'] != 'Unknown')].copy()
4 movies['duration_min'] = movies['duration'].str.replace(' min', '').astype(int)

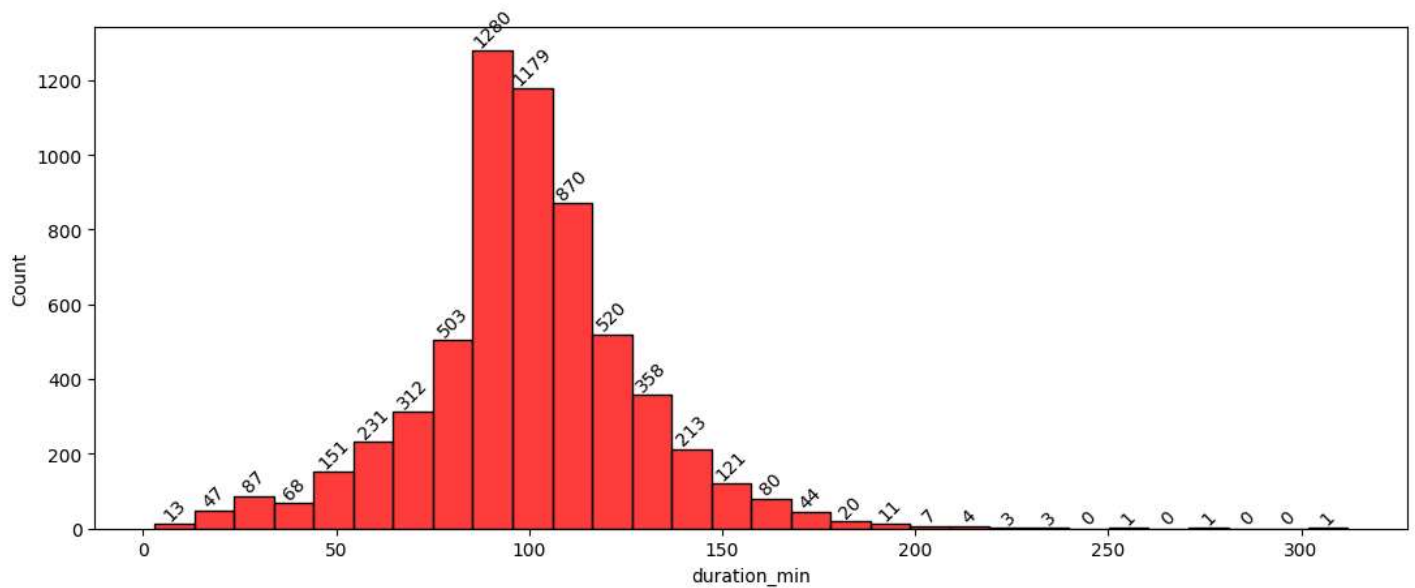
```



```

1 # histogram plot for varying durations
2
3 plt.figure(figsize = (13, 5))
4 mdh = sns.histplot(data = movies['duration_min'], bins = 30, color = 'r')
5 for container in mdh.containers:
6     mdh.bar_label(container, rotation = 45)

```

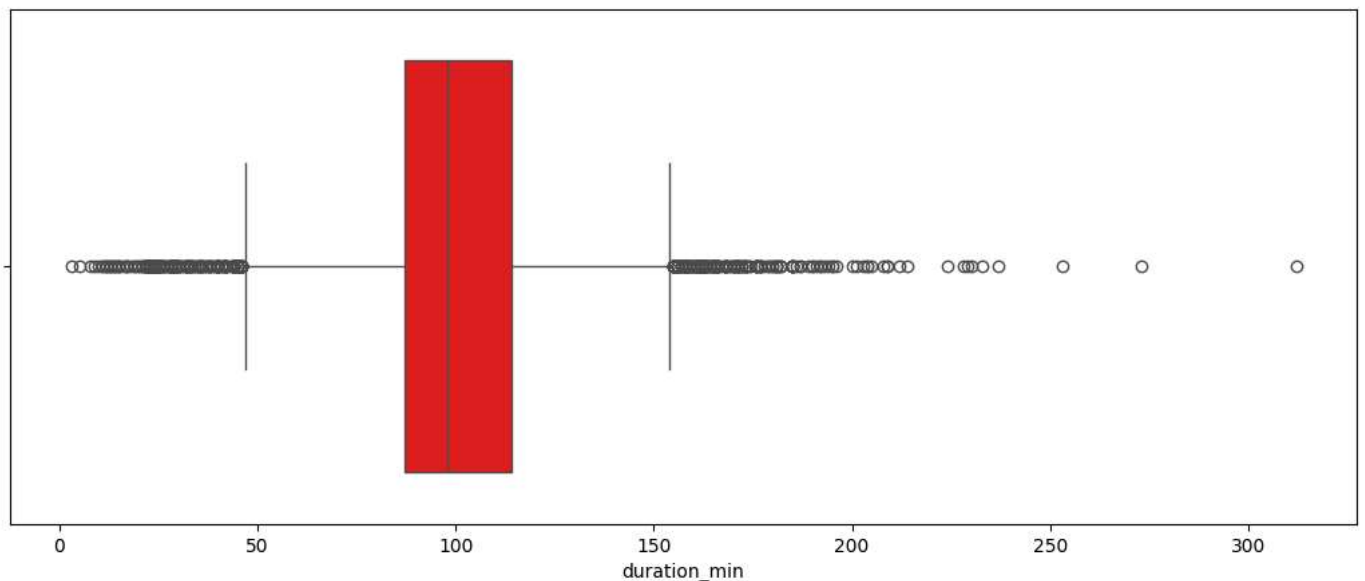


1 # The Histogram plot shows that most of the movies available on Netflix have a duratrion in range 90 - 110 minutes, i,e around 1 hr 3

```

1 # Boxplot for duration statistics for Movies
2
3 plt.figure(figsize = (13, 5))
4 mdb = sns.boxplot(x= movies['duration_min'], color = 'r')

```



```

1 # The boxplot chart shows duration min has an average around 99 min.
2 # The analysis shows multiple movies as outliers in duration_min, which are unique and do not affect the chart data.
3 # The majority of movies have durations close to 100 minutes. A few movies exceed 200 minutes, appearing as outliers.
4 # This indicates that Netflix primarily focuses on standard feature-length movies while maintaining a small collection of exceptiona

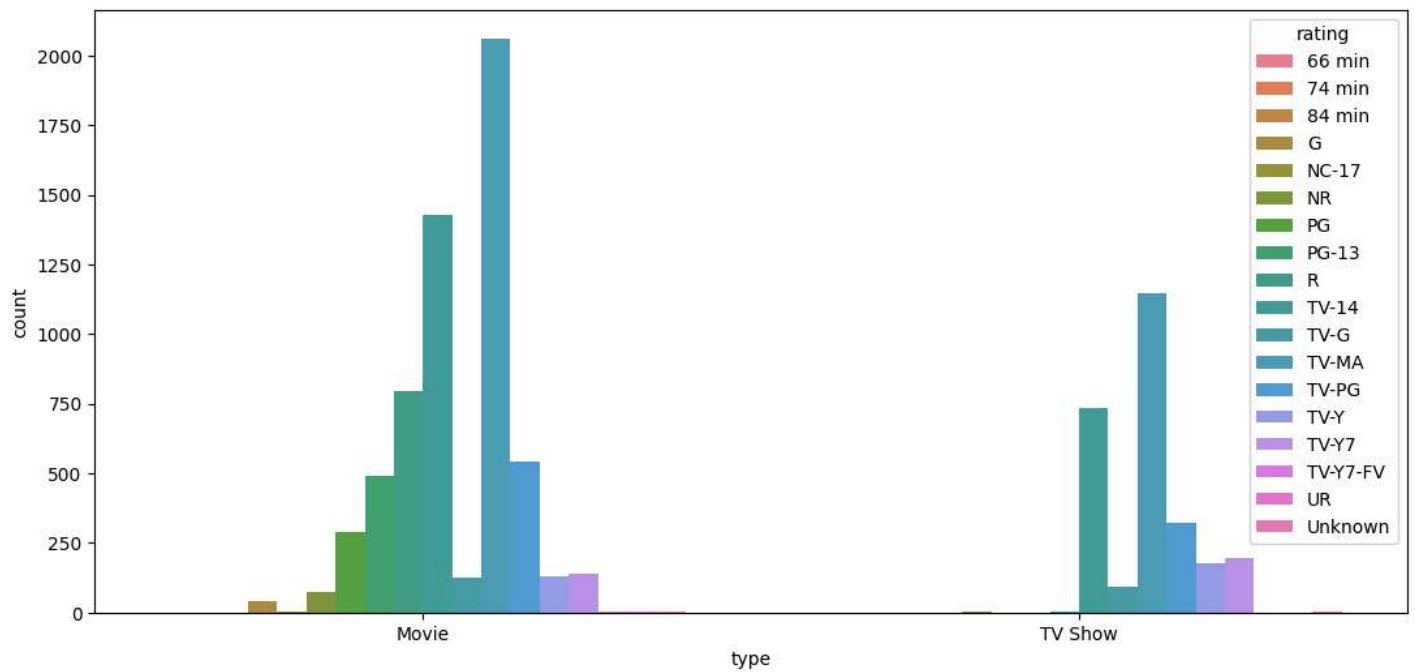
```

Bivariate Analysis

```

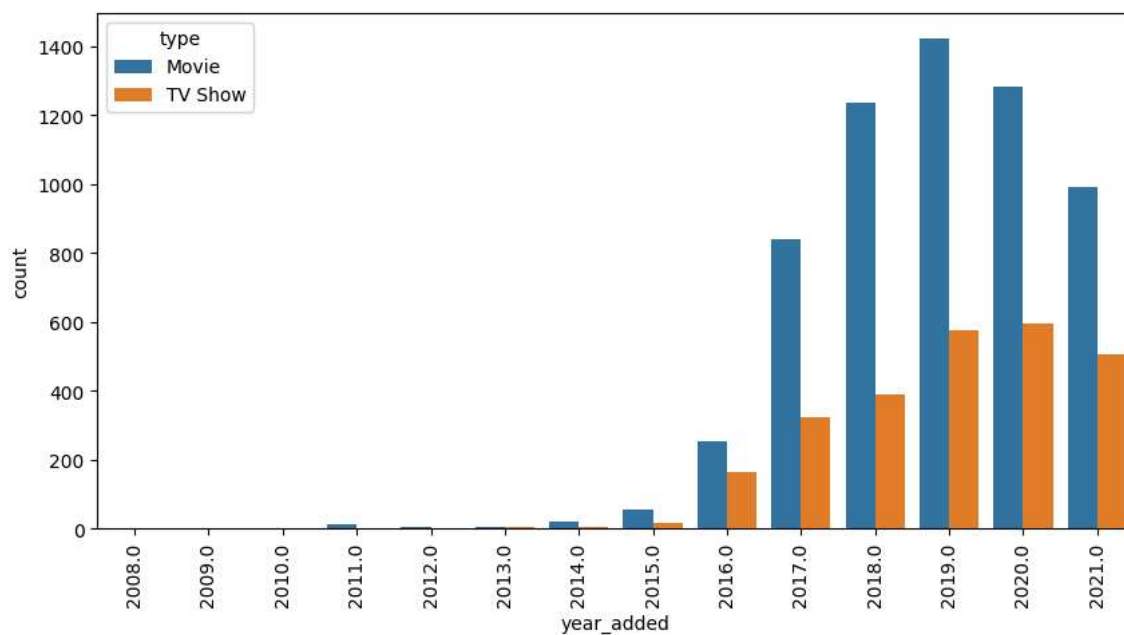
1 # Movieues and Rating relations
2
3 plt.figure(figsize = (13, 6))
4
5 mr = sns.countplot(data = netflix, x = 'type', hue = 'rating')

```



1 # In both type of contents, TV-MA rating tops the charts
 2 # This indicates, Netflix attracting mostly matured viewers

```
1 # Movies, TV Show and their Year Added
2 plt.figure(figsize = (10, 5))
3 mya = sns.countplot(data = netflix, x = 'year_added', hue = 'type')
4
5 plt.xticks(rotation = 90)
6 plt.show()
```



1 # The chart show, most no. of Movies were added in year 2019, where as most no. of TV Show were added in 2020

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```
1 # Top Countries v/s Type
2
3 pd.crosstab(country_n['country'], country_n['type']).sort_values("Movie", ascending=False).head(10)
```

	type	Movie	TV Show
country			
	United States	2752	938
	India	962	84
	United Kingdom	534	272
	Unknown	440	391
	Canada	319	126
	France	303	90
	Germany	182	44
	Spain	171	61
	Japan	119	199
	China	114	48

```
1 # Top Actors and Types
2
3 pd.crosstab(cast_n['cast'],cast_n['type']).sort_values("Movie",ascending=False).head(10)
```

	type	Movie	TV Show
cast			
	Unknown	475	350
	Anupam Kher	42	1
	Shah Rukh Khan	35	0
	Naseeruddin Shah	32	0
	Om Puri	30	0
	Akshay Kumar	30	0
	Amitabh Bachchan	28	0
	Paresh Rawal	28	0
	Julie Teiwani	28	5
	Rupa Bhimani	27	4

```
1 # Top Directors and Types
2
3 pd.crosstab(director_n['director'],director_n['type']).sort_values("Movie",ascending=False).head(10)
```

	type	Movie	TV Show
director			
	Unknown	188	2446
	Rajiv Chilaka	22	0
	Jan Suter	21	0
	Raúl Campos	19	0
	Suhas Kadav	16	0
	Jay Karas	15	0
	Marcus Raboy	15	1
	Cathy Garcia-Molina	13	0
	Martin Scorsese	12	0
	Jay Chapman	12	0

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Overall Business Insights

1. Netflix has more Movies than TV Shows, which means movies are still the main type of content available on the platform.
2. The number of TV Shows has increased in recent years, showing that Netflix is focusing more on series to keep viewers engaged for a longer time.
3. The United States has the highest number of titles on Netflix, followed by countries like India, the United Kingdom, Canada, Japan, and South Korea. This shows that Netflix has a strong presence in these countries.
4. Most of the content available on Netflix was released after the year 2000, which means Netflix mainly offers recent and modern content.
5. Genres like Drama, International Movies, Comedy, and Documentaries are the most common on Netflix. This shows that viewers enjoy these types of content the most.
6. Most movies have a duration of 80 to 120 minutes, which is the standard length for feature films.
7. TV-MA is the most common content rating, which means Netflix mainly targets adult audiences.
8. Netflix works with many different directors and actors instead of depending on only a few people. This helps keep the content library diverse.
9. The number of titles added to Netflix increased rapidly until around 2019, showing that Netflix expanded its content library quickly during that period.
10. Overall, Netflix has a large and diverse content library with content from many countries, genres, and creators, helping it attract viewers from around the world.

Recommendations

1. Netflix should continue investing in TV Shows because they keep viewers engaged for a longer time.
2. Netflix should produce more local content in countries like India, Japan, and South Korea, where regional content is becoming more popular.
3. Netflix should continue making more content in popular genres like Drama, Comedy, and International Movies since these genres have a large audience.
4. Netflix can build long-term partnerships with directors and actors who have contributed several successful titles.
5. Netflix should increase the amount of family and children's content to attract more viewers of all age groups.
6. The company should maintain a good balance between Movies and TV Shows so that different types of viewers can find content they enjoy.
7. Netflix should use data from different countries to understand local audience preferences before producing new content.
8. Netflix should continue expanding its content library with high-quality original content to attract new subscribers and retain existing ones.

✓ Conclusion

- From this analysis, we can conclude that Netflix has a diverse collection of movies and TV shows from different countries and genres. The platform is gradually increasing its focus on TV Shows while continuing to offer a large number of movies. By producing more local content, investing in popular genres, and using audience preferences to guide future decisions, Netflix can continue to grow its business and improve customer satisfaction.

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